‘Just open Chat’ - but what happens to your thinking skills if you use a chatbot often?

In just a few years, the use of generative AI has soared. What happens to your thinking skills when you constantly turn to a chatbot? ‘Learning takes effort. Chatbots try to remove that friction.’

Laurens Verhagen and Kaya Bouma | de Volkskrant | 29 October 2026, 18:00

He wholeheartedly recommends it to his son whenever the 16-year-old needs to memorise vocabulary or has a week of tests coming up. ‘Use ChatGPT,’ Stefan van der Stigchel tells him. ‘If you can no longer concentrate or you get stuck, just open Chat and let it ask you a few questions.’

Endlessly patient and always available: AI can be an excellent teacher, believes the Utrecht professor of cognitive psychology. ‘AI can offer a real opportunity to people who have no access to education, and to children who do not have patient parents at home to help them with their homework.’

Van der Stigchel himself researches concentration. While social media mainly distract pupils from their homework, he expects AI to have the opposite effect. ‘Having TikTok on your phone makes it harder to focus, but a chatbot can help you concentrate on something for longer and in greater depth. Provided you use it in the right way.’

Pen and paper

Anco Peeters is unlikely to encourage his students to use ChatGPT. Quite the opposite: during the philosopher and cognitive scientist’s seminars at Radboud University in Nijmegen, students have to close their laptops and put their phones away.

During those sessions, they must work on an essay entirely under their own steam. First by discussing it with each other, then by writing. With pen and paper. No Google, no ChatGPT: they have to think for themselves about good questions and how to structure an argument. That, Peeters believes, is how his students learn the most.

Two academics working on thinking skills and AI, and two completely different approaches. It says a great deal about the state of science and education when it comes to using AI wisely. While schools and universities rush to adapt their teaching to the AI age, the fundamental question beneath it all remains unanswered.

‘A golden age of stupidity’

What does AI do to our thinking skills? Is it the ideal teacher, helping us become smarter? Or does AI make us less intelligent because we outsource our thinking while immersing ourselves in an endless stream of mind-numbing videos? The British newspaper *The Guardian* has already coined the phrase ‘a golden age of stupidity’.

Such questions are becoming increasingly urgent as generative AI becomes more deeply woven into everyday life. Around 65 per cent of Dutch adults use generative AI, mainly chatbots such as ChatGPT or Claude. Among 18- to 35-year-olds, the figure is as high as 89 per cent, according to research by the University of Twente published earlier this year.

Experts argue that this widespread use, achieved in just a few years, is not without danger. If students or other members of the public leave some of their thinking to chatbots, they risk losing critical skills.

Writing is thinking

The British-Australian cognitive philosopher Richard Menary put it very simply: writing is thinking. To put it bluntly, if chatbots take over this work, we think less for ourselves and, over the longer term, develop ‘brain rot’, the popular term that quickly crops up in this discussion.

In an earlier interview with *de Volkskrant*, psychologist and author Sanne Cornelissen described the vicious circle people can find themselves in: ‘Because you think less, you stop training this ability and become less confident in your own capabilities. That makes you outsource your work to AI even more often, and it becomes increasingly difficult to think critically and express yourself independently.’

This makes people increasingly dependent on chatbots, and therefore on technology companies, warns cognitive scientist Peeters. ‘Companies such as OpenAI have a very clear interest in making people dependent on them for their thinking. That is their business model. At some point, no one will be able to go back.’

The consequences could be far-reaching. If citizens are no longer educated to be critical thinkers, that also undermines the foundations of a healthy democracy. ‘If automated reasoning displaces our own capacities for moral and political thought, it leads to what I call “moral deskilling”: the moral equivalent of a world in which no one can navigate with a map any more because we are entirely dependent on GPS,’ writes Shannon Vallor, a Scotland-based professor of ethics and AI, in her book *The AI Mirror*.

Socrates

When it comes to navigation, we may be willing to accept that trade-off, she argues. ‘But should we accept the same bargain when it comes to outsourcing our capacity for moral and political deliberation? What would be left of us then?’

Existential questions like these are nothing new. The original argument against technology that impairs our thinking comes from the Greek philosopher Plato. Or perhaps from his teacher Socrates, who appears in Plato’s dialogue *Phaedrus* (fourth century BC) in a discussion about writing. Writing, Socrates argues, weakens memory because people rely on external symbols. The irony: Socrates himself left no writings behind; his pupil Plato did. Through writing, of course.

Similar objections were heard with the arrival of printing. For monks, copying texts was not merely an inefficient way of reproducing them, but also a form of study, discipline and devotion. The Dominican friar Filippo de Strata compared the then-modern printing press to a ‘whore’. The pen was a virgin.

More irony: in an interview last year, OpenAI chief executive Sam Altman let slip that he likes to reach for a pen rather than a keyboard when he really needs to think.

The calculator

Another frequently cited example is the calculator, which has made people less proficient at mental arithmetic. Yet nobody wants to return to the days before calculators, enthusiastic chatbot users often point out. The subtext: why make a fuss about ChatGPT and its peers?

That comparison is flawed, says Peeters, because generative AI can take over the entire thinking process: ‘The calculator has a limited cognitive scope.’ Laura Koenders, an educational adviser at Utrecht University, also cautions against drawing parallels with earlier technologies such as navigation apps or calculators. ‘AI also involves language, and therefore thinking. There are not many situations you can imagine in which you would want to outsource that part,’ Koenders says.

What does research tell us about cognitive decline resulting from chatbot use? The best-known study is an experiment at the Massachusetts Institute of Technology. It is a small-scale study and, moreover, a preprint: the research has not yet been peer-reviewed. But the results were so striking that journalists around the world clamoured to speak to lead researcher Nataliya Kosmyna when she posted an initial version online in June 2025.

In the experiment, Kosmyna and her colleagues asked 54 participants, mainly students, to write essays on several occasions. One group had to do so entirely unaided; a second could use the internet but not AI; and a third was given full access to ChatGPT.

Brain activity

The researchers monitored participants’ brain activity using EEG while they wrote their essays. Activity was strongest and most widespread in the group that wrote without help. Brain regions associated with creative idea generation, reading and writing were particularly active. The AI users had the lowest levels of brain activity.

Remarkably, those differences persisted when the researchers reversed the roles and the group that had started with AI then had to write an essay entirely on their own. Their brain activity remained relatively low, as though their brains had slipped into a kind of sleep mode.

That sounds dramatic, but Peeters offers a critical caveat. A brain scan can easily be mistaken for an indisputable biological fact. Just look at the scan: we are getting less intelligent! But brain scans do not tell us much on their own, he stresses. ‘You can see that different connections are being made in the brain, but that does not immediately mean it is harmful. You need behavioural research alongside it.’

In other words, a reduction in certain brain waves or network connections does not necessarily mean that the brain is no longer working. It may mean that participants are using a different cognitive strategy: evaluating and filtering AI suggestions, for example, rather than retrieving information from their own memory.

Seen in that light, what happened after the participants in the American study submitted their essays is more telling. When asked, none of the AI users could reproduce even one correct sentence from the work they had just handed in. The other two groups had little difficulty doing so.

Cognitive debt

The researchers use the term ‘cognitive debt’. It arises when people consistently outsource effortful thinking processes. In the short term, this offers convenience and saves time. Over the longer term, however, the enthusiastic ChatGPT user builds up a debt in the form of diminished critical thinking skills, lower creativity or poorer memory.

A large-scale study published this summer also paints an unfavourable picture of what AI does to thinking. The researchers followed more than 26,000 secondary-school pupils in China for 30 months. At the start of the study period, in September 2022, none of them was using AI. Over the following two and a half years, more and more pupils began using AI tools. There was also a control group that did not use AI for homework at all.

The AI users soon achieved higher homework marks while spending substantially less time on it. So far, so good. But when pupils had to take exams in class, their results fell dramatically: by an average of 20 per cent within six months.

By avoiding mental effort, they had developed a cognitive deficit, the researchers conclude. Meanwhile, pupils who

continued spending the same amount of time on homework and used AI only as support experienced hardly any learning loss.

A chatbot can help, too

There are also studies showing that AI really can help students, or at least does them no harm. The condition is that the chatbot must be used very deliberately. In a large-scale experiment involving secondary-school pupils, Hamsa Bastani of the University of Pennsylvania compared a specially developed ‘AI tutor’ with the standard version of ChatGPT.

This tutor came with educational training wheels: it did not provide direct answers, but asked targeted questions and offered hints based on common mistakes. On practice exercises where pupils received AI assistance, both groups performed better than pupils who used no AI. Standard ChatGPT improved performance by 48 per cent; with the AI tutor, the improvement was as much as 127 per cent.

The big difference emerged in exams. ChatGPT users came unstuck: their scores fell by an average of 17 per cent compared with the control group without AI help. AI tutor users performed at a comparable level to the control group. All AI users overestimated the skills they had learnt.

It is a familiar pitfall, says Peeters: ‘When you have support in the form of a chatbot, you overestimate your own abilities. You soon think: oh yes, that answer is exactly what I had in mind myself.’ Peeters compares it to looking up the answers at the back of a maths textbook too soon. ‘There too, you soon think you would have come up with it yourself.’

The AI tutor

In another study, involving Harvard students, assistance from an AI tutor produced even better results. All the students were taking the same physics course. One group attended an active classroom physics lesson; the other studied at home with a specially developed AI tutor.

The students taught with AI learnt more in less time, according to tests taken before and afterwards. The researchers attributed those positive results mainly to the individual AI tutor’s tailored teaching: students received feedback and explanations suited to their level. A human tutor would probably achieve the same result, the researchers said.

The existing research does not provide an unequivocal answer to whether AI makes us less intelligent, Professor Van der Stigchel concludes. ‘The unsatisfying scientific answer is: sometimes it does and sometimes it does not. It depends on how you use AI. If you use it to explore a subject in depth, or have AI rigorously test your knowledge, you actually learn from it.’

That, Van der Stigchel says, is the crux: the mental effort we put in. When our brains have to work hard, knowledge sticks better. ‘If you use ChatGPT to explain a particular maths problem and then practise for yourself, you learn. But if you let Chat do the problems for you, you learn nothing.’

The shopping list

A term that often comes up in this discussion is ‘cognitive offloading’: delegating thinking to something else, something outside ourselves. There is not necessarily anything wrong with that, says Koenders. Take something as simple as a shopping list or a to-do list. This kind of outsourcing frees up mental space for other, more complex matters, such as problem-solving.

And sometimes writing things down can itself contribute to better understanding: taking written notes during a lecture forces the brain to summarise and restructure information.

But, like Van der Stigchel, Koenders and Peeters warn against thoughtlessly outsourcing thinking to chatbots that keep churning out answers. Koenders: ‘Learning takes effort. You have to keep revisiting the material; you have to make an effort. That is all part of the process. The companies behind commercial chatbots are doing everything they can to remove that friction.’ ChatGPT then becomes a very tempting shortcut: ‘Many students have a lot of balls to keep in the air, or very little time. Almost no one can resist it.’

A divide

Specially designed AI tutors could solve this problem. Peeters also envisages what he calls ‘reflection machines’: ‘You can use AI systems to ask Socratic questions. For doctors, for example, or lawyers, too. A machine like that does not give answers; it asks questions. For instance: have you considered this alternative explanation? And if so, why did you reject it?’ It is not necessarily a pleasant experience, Peeters admits: ‘You have to create friction yourself, for example by spelling out the basis on which you are making a decision.’

Using AI in this way requires knowledge and discipline. Could a divide eventually emerge between users who deploy AI thoughtlessly and lose cognitive ground, and those who use chatbots very deliberately and become smarter as a result?

That is already happening, Professor Van der Stigchel believes. ‘There is already a group using AI incorrectly. That is why it is so important to improve people’s understanding of how to use it wisely. Schools have a role here: tell pupils how to use it intelligently, because the fact is that they are using it. But I hope adults read this article too, and start thinking about it.’

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